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Class : SYB(68)

AIM: To implement and analyze Binomial, Poisson, and Normal distributions using Python.

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📘 Q1. What is the difference between Binomial, Poisson, and Normal distributions?

Answer:

Binomial, Poisson, and Normal distributions are important probability distributions used to model different types of data.

The Binomial distribution is a discrete distribution used when there is a fixed number of trials and each trial has only two possible outcomes: success or failure. It depends on two parameters: the number of trials (n) and the probability of success (p). For example, counting the number of heads in coin tosses.

The Poisson distribution is also a discrete distribution but is used to model the number of rare events occurring in a fixed interval of time or space. It does not require a fixed number of trials. It is defined by a single parameter λ (lambda), which represents the average rate of occurrence. For example, number of calls received per hour.

The Normal distribution is a continuous distribution and has a bell-shaped curve. It is symmetric about the mean and is defined by mean (μ) and standard deviation (σ). It is used for continuous data such as height, weight, and marks.

Thus, Binomial is used for fixed trials, Poisson for rare events, and Normal for continuous data.

📘 Q2. What is the real-world significance of the Poisson distribution?

Answer:

The Poisson distribution is highly significant in real-world applications because it is used to model rare and random events that occur over a fixed period of time or space.

One of its main uses is in resource planning, where organizations estimate the number of events such as customer arrivals or service requests. This helps in allocating staff and reducing waiting time.

It is also important in risk management, where it helps calculate the probability of rare but critical events such as system failures, accidents, or breakdowns.

In queueing theory, the Poisson distribution is used to model arrival rates in systems like banks, hospitals, and call centers. This helps in designing efficient systems and minimizing delays.

In quality control, industries use it to measure defects in products and maintain standards.

In web analytics, it helps analyze traffic, clicks, or user activity over time.

Therefore, the Poisson distribution is widely used for prediction, planning, and decision-making in many fields.

📘 Q3. What is the Central Limit Theorem and how does it relate to Normal distribution?

Answer:

The Central Limit Theorem (CLT) is one of the most important concepts in statistics. It states that if we take sufficiently large random samples from any population, the distribution of the sample means will be approximately normal, regardless of the shape of the original population distribution.

According to CLT:

The mean of the sample means is equal to the population mean (μ). The standard error is equal to σ / √n, where σ is the population standard deviation and n is the sample size.

The relationship between CLT and the Normal distribution is very important. Even if the original data is skewed or irregular, the distribution of sample means becomes bell-shaped (normal) when the sample size is large (generally n ≥ 30).

This theorem explains why the Normal distribution appears frequently in real-world data and is widely used in statistical analysis.

📘 Q4. Why is Normal distribution widely used in machine learning and statistics?

Answer:

The Normal distribution is widely used in statistics and machine learning due to its simplicity, natural occurrence, and strong mathematical properties.

Many real-world phenomena such as height, weight, IQ, and measurement errors naturally follow a Normal distribution. This makes it very useful for modeling real data.

It is defined using only two parameters: mean (μ) and standard deviation (σ), which makes it simple and easy to analyze.

The Central Limit Theorem further supports its importance by stating that the distribution of sample means tends to be normal even if the original data is not.

Most statistical techniques such as hypothesis testing, confidence intervals, and regression analysis are based on the assumption of Normal distribution.

In machine learning, many algorithms like Linear Regression, Gaussian Naive Bayes, and probabilistic models assume that the data follows a Normal distribution.

Additionally, the empirical rule (68–95–99.7 rule) helps in understanding the spread of data easily.

Thus, the Normal distribution is widely used because it is simple, practical, and mathematically powerful.

📘 Q5. What is skewness in distributions and which distributions exhibit it?

Answer:

Skewness is a statistical measure that describes the asymmetry of a probability distribution around its mean. It indicates whether the data is evenly distributed or tilted to one side.

There are three main types of skewness:

1. Positive Skewness (Right-Skewed Distribution): In this case, the tail of the distribution is longer on the right side. A few large values pull the mean towards the right. The relationship is: Mean > Median > Mode. Example: Income distribution where a few people earn very high salaries.
2. Negative Skewness (Left-Skewed Distribution): Here, the tail is longer on the left side. A few small values pull the mean towards the left. The relationship is: Mean < Median < Mode. Example: Marks in an easy exam where most students score high.
3. Zero Skewness (Symmetrical Distribution): In this case, the distribution is perfectly symmetrical, and the mean, median, and mode are equal. Example: Normal distribution.

Mathematically, skewness can be measured using the Fisher-Pearson coefficient: $\gamma_1 = E[(X - \mu)^3] / \sigma^3$

Different distributions exhibit different skewness:

Normal distribution → Zero skewness Poisson distribution → Usually positively skewed Binomial distribution → Can be positively skewed, negatively skewed, or symmetric depending on the value of p

Skewness is important because it helps in understanding the shape of data, selecting appropriate statistical methods, and improving data analysis and machine learning models.

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

df.head()
```

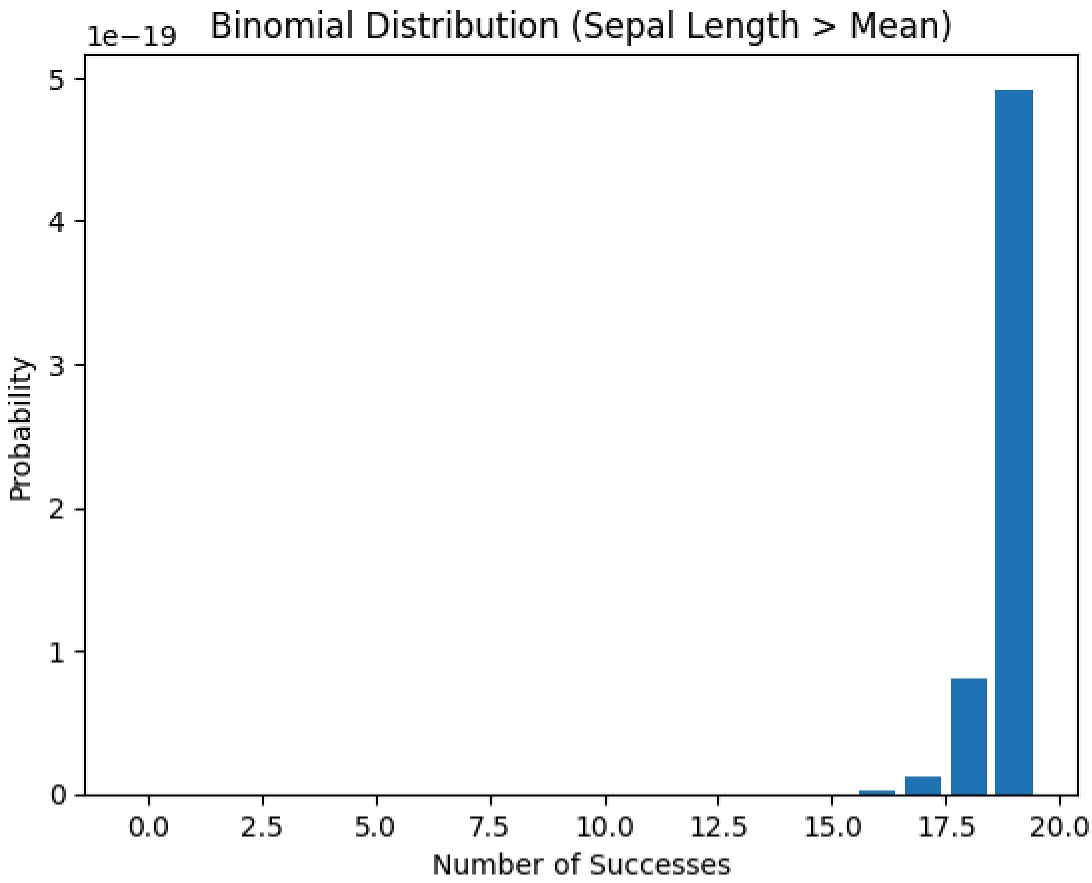
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
# Create binary data
mean_val = df['sepal length (cm)'].mean()
df['success'] = df['sepal length (cm)'] > mean_val

# Count successes
n = len(df)
k = df['success'].sum()
p = k / n

# Binomial PMF
x = np.arange(0, n+1)
y = binom.pmf(x, n, p)

# Plot (limited range for visibility)
plt.figure()
plt.bar(x[:20], y[:20])
plt.title("Binomial Distribution (Sepal Length > Mean)")
plt.xlabel("Number of Successes")
plt.ylabel("Probability")
plt.show()
```

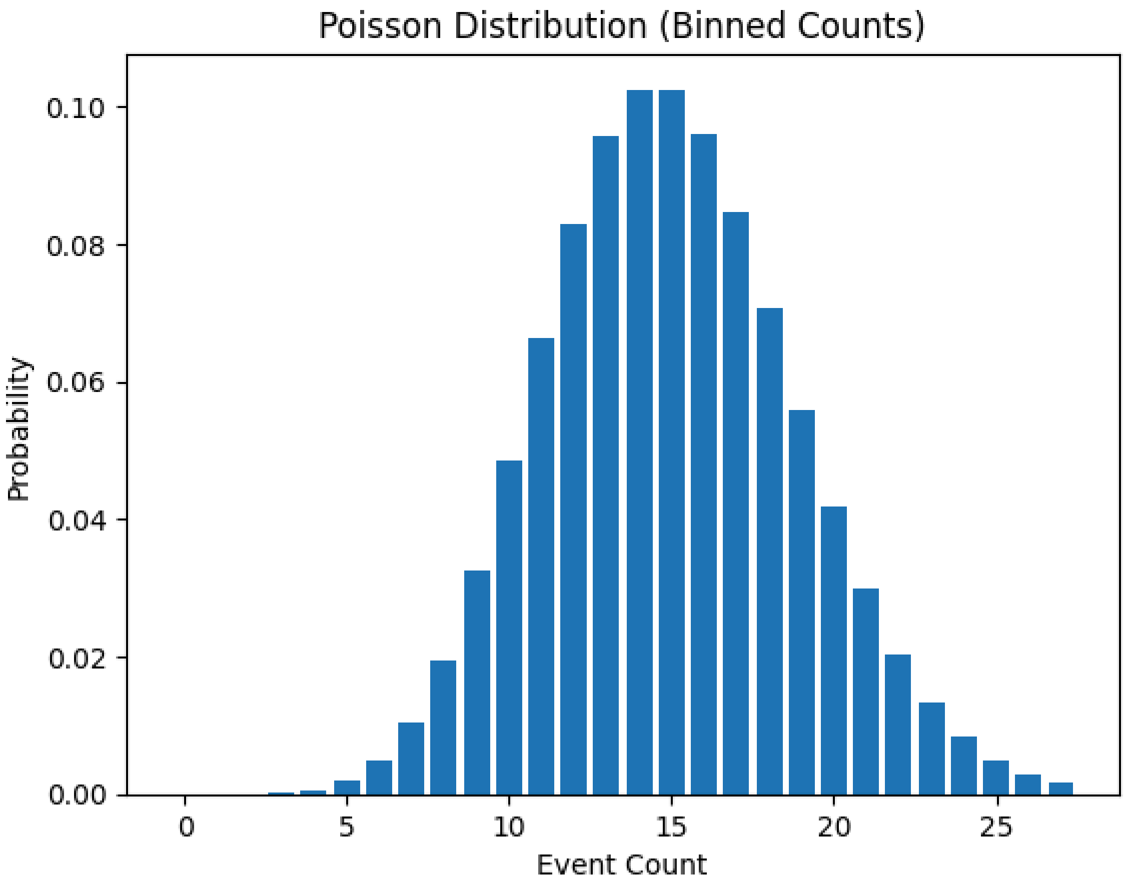



```
# Count values per bin (simulate event counts)
counts, _ = np.histogram(df['sepal length (cm)'], bins=10)

# Mean rate
lam = np.mean(counts)

# Poisson PMF
x = np.arange(0, max(counts)+1)
y = poisson.pmf(x, lam)

# Plot
plt.figure()
plt.bar(x, y)
plt.title("Poisson Distribution (Binned Counts)")
plt.xlabel("Event Count")
plt.ylabel("Probability")
plt.show()
```

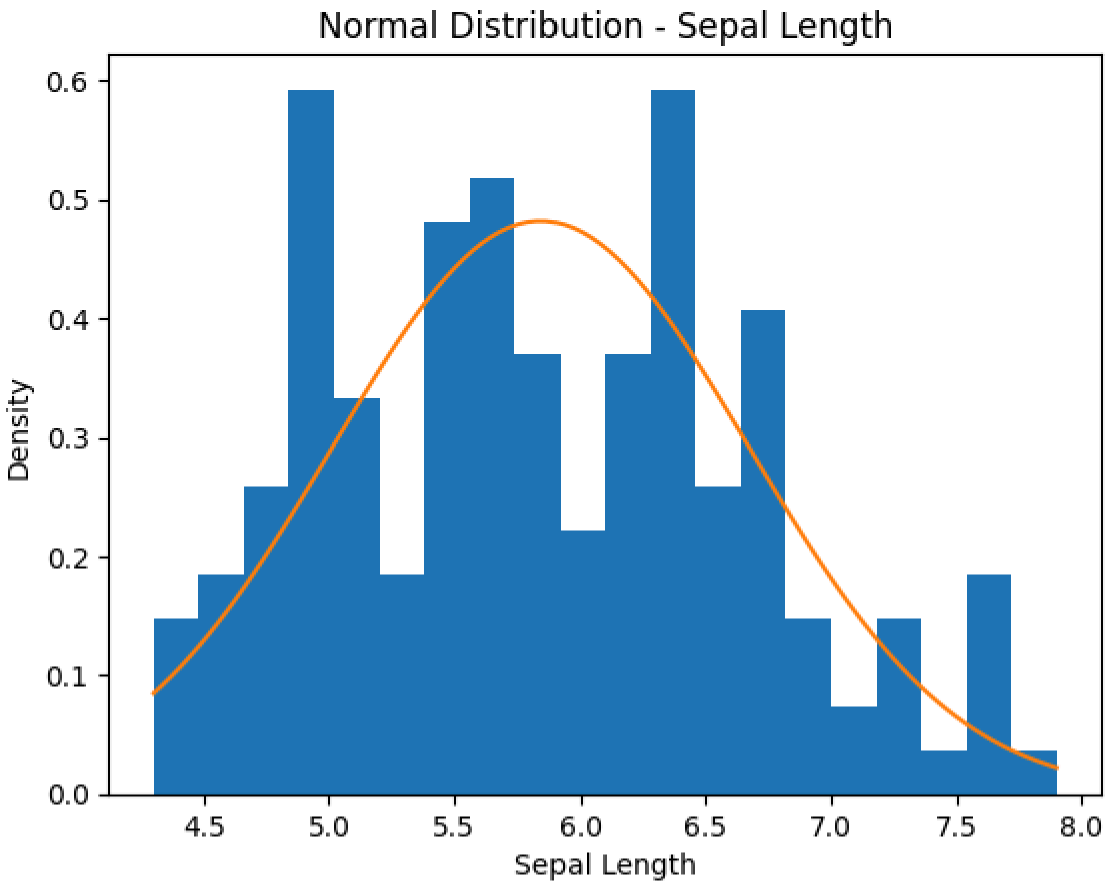



```
# Use sepal length (continuous data)
data = df['sepal length (cm)']

# Parameters
mu = data.mean()
sigma = data.std()

# Generate values
x = np.linspace(data.min(), data.max(), 100)
y = norm.pdf(x, mu, sigma)

# Plot
plt.figure()
plt.hist(data, bins=20, density=True)
plt.plot(x, y)
plt.title("Normal Distribution - Sepal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Density")
plt.show()
```




```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# Load dataset is handled in a previous cell using sklearn.datasets.load_ir
# df = pd.read_csv('/content/iris[1].csv') # Removed this line

# 1. NORMAL DISTRIBUTION (Direct)
data = df['sepal length (cm)']
mu = data.mean()
sigma = data.std()

x_norm = np.linspace(data.min(), data.max(), 100)
y_norm = norm.pdf(x_norm, mu, sigma)

# ----------------------------
# 2. BINOMIAL DISTRIBUTION
# ----------------------------
mean_val = data.mean()
df['success'] = data > mean_val

n = len(df)
p = df['success'].sum() / n

x_bin = np.arange(0, 20)  # limited for visibility
y_bin = binom.pmf(x_bin, n, p)
```

```
# ----------------------------
# 3. POISSON DISTRIBUTION
# ----------------------------
counts, _ = np.histogram(data, bins=10)
lam = np.mean(counts)

x_pois = np.arange(0, max(counts)+1)
y_pois = poisson.pmf(x_pois, lam)

# ----------------------------
# PLOTTING COMPARISON
# ----------------------------
plt.figure()

# Normal
plt.plot(x_norm, y_norm, label='Normal Distribution')

# Binomial
plt.bar(x_bin, y_bin, alpha=0.5, label='Binomial')

# Poisson
plt.bar(x_pois, y_pois, alpha=0.5, label='Poisson')

plt.title("Comparison of Distributions on Iris Dataset")
plt.xlabel("Values / Counts")
plt.ylabel("Probability / Density")
plt.legend()
plt.show()
```